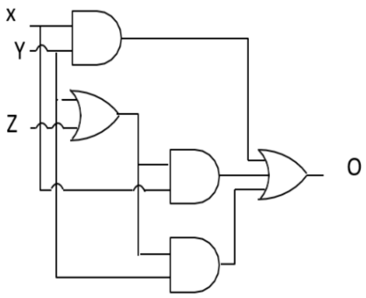
Digital electronics question

1. What is the minimum number of gates required to implement the Boolean function (AB+C) if we have to use only 2-input NOR gates?
   (a) 2 (b) 3 (c) 4 (d) 5
2. Design AND, OR, NOT, Xor and XNOR gates using NAND or NOR gates
3. Consider the following Boolean expression for F:
   F (P, Q, R, S) = PQ + P'QR +P'QR'S The minimal sum-of-products form of F is

   (a) **PQ + QR + QS** (b) P + Q + R + S (c) P' + Q' + R' + S' (d)P'R + P'R'S + P

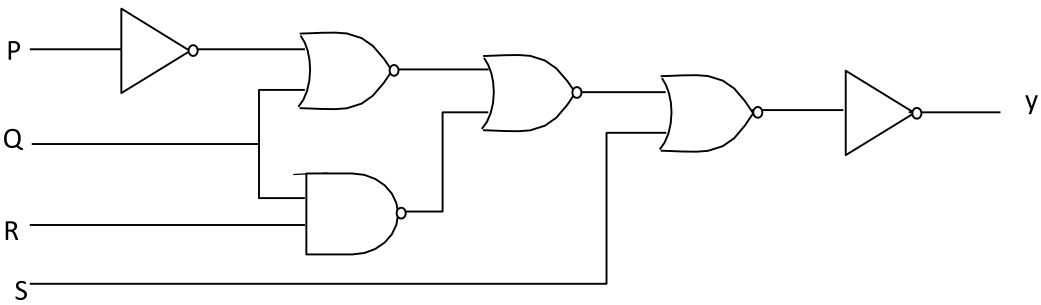
4. The Boolean expression $(\overline{AB})(\overline{A}+B)(A+\overline{B})$ can be simplified to
   (a)A +B (b) $\overline{A}B$ (c) $\overline{A+B}$ (d) AB

5. The output of following logic circuit can be simplified to



   (a) X+YZ (b) Y + ZX (c) XYZ (d) X + Y +Z

6. The logic expression for the output Y of the following circuit is



(a) $\overline{\overline{\overline{P}+Q}+QR+\overline{S}}$ (b) $\overline{\overline{\overline{P}+Q}+Q\overline{R}+S}$ (c) $\overline{\overline{P}+Q+QR+\overline{S}}$ (d) None

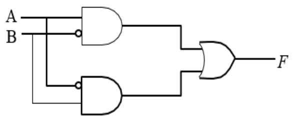
7. Which one of the following is an INCORRECT Boolean expression?
   (a) $\overline{P}Q + PQ = Q$ (b) $(\overline{P} + Q)(P + Q) = P$ (c) P (P+Q) = P (d) None

8. Consider the following truth table The logic expression for *F* is
   (a) $AB + BC + CA$ (b) $\overline{A}B + A\overline{C} + \overline{B}C$ (c) $\overline{C\overline{AB}} + A\overline{B}$ (d) $\overline{C}(A + \overline{B}) + A\overline{B}$

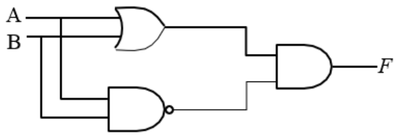
A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

9. Which of the following circuit does not satisfy the Boolean expression $A\overline{B} + \overline{A}B = F$

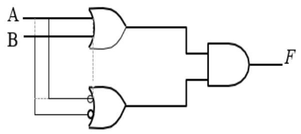
(a)



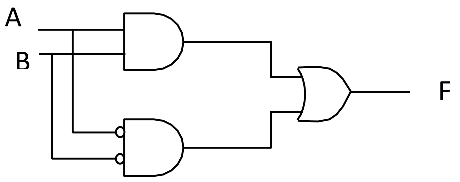
b



(c)



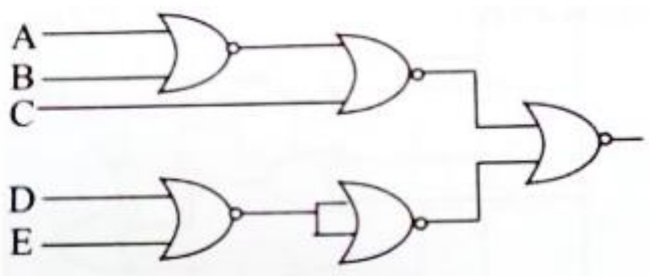
(d)



10. The Boolean expression $P + \overline{P}Q$, where P and Q are the inputs to a circuit, represents
    (a) AND (b) NAND (c) NOT (d) OR
11. Octal equivalent of decimal number $(478)_2$ is
    (a) 736_8 (b) 673_8 (c) 637_8 (d) 567_8
12. conversion from one form to another

    i. $(239.45)_{10} = (\quad)_8$
    ii. $(6327.4051)_8 = (\quad)_{10}$
    iii. $(45.625)_{10} = (\quad)_2$
    iv. $(123.6875)_{10} = (\quad)_2$
    v. $(11011.101)_2 = (\quad)_{10}$
    vi. $(101100.10101)_2 = (\quad)_{10}$
    i. $(234.45)_{10} = (\quad)_{16}$
    ii. $(EA4.45)_{16} = (\quad)_{10}$
    iii. $(234.45)_8 = (\quad)_2$
    iv. $(234.45)_8 = (\quad)_{16}$
    v. $AB4.45)_{16} = (\quad)_2$
    vi. $(AB4.45)_{16} = (\quad)_8$
13. Subtract two binary numbers using 1's complement and 2's complement method.
14. Subtract two decimal numbers using 9's complement and 10's complement method
15. Subtract two octal numbers using 7's complement and 8's complement method
16. Subtract two hexadecimal numbers using 15's complement and 16's complement method
17. Implementation of following Boolean function using only universal gates (NAND/ NOR) gates
    i. F(A, B, C) = A' + BC'
    ii. F(A, B, C, D) = A + A'B (C'+D)
18. Expand $A(\overline{B} + A)B$ in both POS and SOP form

19. Reduce the expression $\prod \mathbf{M}$ (2, 8, 9, 10, 11, 12, 14) in both SOP and POS form using K-Map and implement it using NOR gates assuming that the complemented inputs are already given.
20. Reduce the expression $\sum \boldsymbol{m}$ (1, 5, 6, 12, 13, 14) + d(2,4) in both SOP and POS form using Kmap
21. Determine the minimum number of NAND gates required to implement: f = A + A$\bar{\mathbf{B}}$+ A$\bar{\mathbf{B}}$C
22. Determine the output expression of the following circuit:



23. Find the minimized expression for the given K-maps:

CD \ AB	00	01	11	10
00		1	1	
01		1	X	
11	1	1	X	X
10	1		X	X

WX \ YZ	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		